

Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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The Chemical Master Equation: exact stochastic distributions

A well-mixed system: integer counts $\mathbf{n} \in \mathbb{Z}_{\geq 0}^S$ evolving by stochastic reactions.

The probability $p(\mathbf{n}, t)$ satisfies the **CME**:

$$\dot{p} = A p,$$

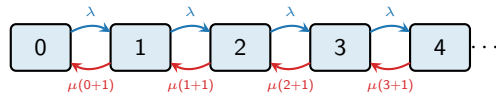
where A encodes every reaction event as a transition rate.

SSA vs. CME

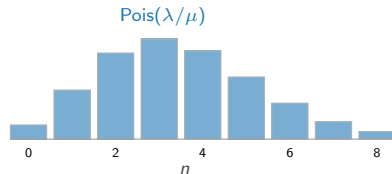
SSA samples individual trajectories.

CME gives the *exact* joint distribution $p(\mathbf{n}, t)$ directly — rare events, tails, and multimodal structure without ensemble noise.

Birth-death: $\emptyset \xrightarrow{\lambda} X \xrightarrow{\mu n} \emptyset$



Stationary distribution: $\text{Poisson}(\lambda/\mu)$.



FSP: truncate A to a finite support $\mathcal{S}$ and solve $\dot{p} = A p$

The RDME: reactions and diffusion on a spatial grid

Divide a 2-D domain into $K \times K$ voxels. Each voxel $\mathbf{k}$ holds a molecule count $n_{\mathbf{k}} \in \mathbb{Z}_{\geq 0}$.

Two types of events:

- **Reactions** inside voxel $\mathbf{k}$: propensity $\alpha(n_{\mathbf{k}})$.
- **Diffusion** between neighbors $\mathbf{k} \leftrightarrow \mathbf{l}$: rate $d = D/h^2$, moves one molecule.

The joint probability $p(\{n_{\mathbf{k}}\}, t)$ satisfies $\dot{p} = Ap$ (**RDME**).

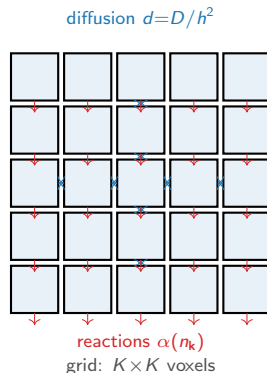
The curse of dimensionality

Joint state space: $\sim n_{\max}^{K^2}$ entries.

A 200×200 grid is **completely intractable** directly.

Key locality: diffusion only couples *adjacent* voxels.

This structure enables a principled decomposition.



Per-voxel CME with mean-field diffusion coupling

Key idea: diffusion couples voxels through their means.

A molecule hops from voxel ℓ to voxel k at rate $d \cdot n_\ell$. Replace n_ℓ by its mean $\mu_\ell = \langle n_\ell \rangle$:

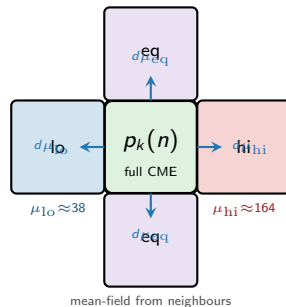
Mean-field RDME decomposition

Diffusion in voxel k : birth at rate $d \sum_{\ell \sim k} \mu_\ell$
death at rate $d \cdot |\partial k| \cdot n_k$

Each voxel k evolves under an **effective 1-D CME**:

$$\dot{p}_k = Q_k(\{\mu_\ell\}_{\ell \sim k}) p_k$$

Why it works: at the wavefront, equil voxels are sharp (delta-like at their basin), so μ_ℓ fully characterises them. The ADI pair step handles active-active coupling.



Adaptive resolution: active vs. equilibrated voxels

Active voxels (wavefront)

Full per-voxel CME via $\expv(dt, Q_k, p_k)$.

Tridiagonal generator Q_k — $O(w)$ Krylov cost where $w \approx 20$ – 50 is the active support width.

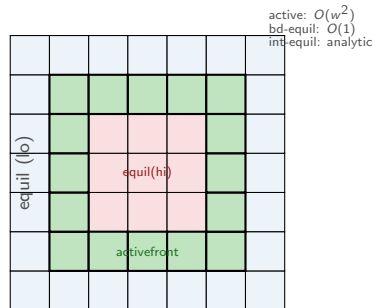
Equilibrated (equil) voxels

Distribution locked near a stable basin.

Boundary equil (adjacent to active): $\expv$ at reduced Krylov dimension $m_{\text{eq}}=8$.

Interior equil: algebraic update only —
 $\mu_k \leftarrow (k_b + \text{in}) / (k_d + \text{out})$, $p_k \leftarrow \text{Pois}(\mu_k)$.
No matrix exponential needed.

ADI pair step (active-active pairs)



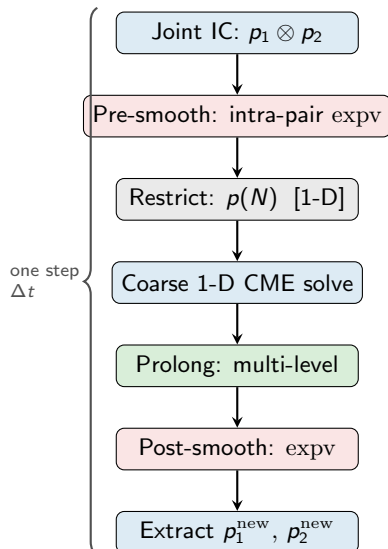
At peak: BD 200×200 grid — ~ 5000 active voxels (thin ring), ~ 35000 equil (interior), ~ 0 equil (exterior, not yet reached).

Interior equil updated analytically: $O(1)$ per voxel.

The ADI pair V-cycle: exact 2-voxel RDME at the front

For each adjacent active pair (v_1, v_2) :

- 1 Build sparse joint IC: $p_1 \otimes p_2$ (product state)
- 2 **Pre-smooth**: evolve intra-pair diffusion $v_1 \leftrightarrow v_2$ for $\tau_{\text{pre}} = 0.1 \Delta t$
- 3 **Restrict**: marginalise to total count $p(N) = \sum_{n_1+n_2=N} p(n_1, n_2)$ [1-D vector]
- 4 **Coarse solve**: 1-D CME on N with external boundary flux, for full Δt
- 5 **Prolong**: multi-level reconstruction of joint $p_{\text{new}}(n_1, n_2)$ from $p_{\text{post}}(N)$
- 6 **Post-smooth**: intra-pair diffusion for τ_{post}
- 7 Extract marginals $p_1^{\text{new}}, p_2^{\text{new}}$



ADI alternation

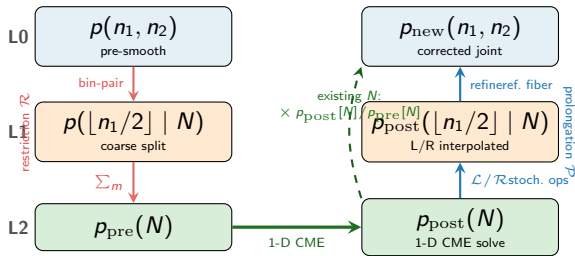
Multi-level prolongation: three-level hierarchy

The restriction erases the split.

Marginalising $p(n_1, n_2)$ to $p(N)$ loses all information about *how* N molecules are distributed across v_1 and v_2 .

Three-level hierarchy on the split axis:

- **L0 (fine):** $p(n_1 | N)$ exact conditional
- **L1 (mid):** $p(\lfloor n_1/2 \rfloor | N)$ — pair consecutive n_1 into bins of width 2
- **L2 (coarse):** $p(N)$ total count only



Prolongation rule

Existing N (in support of p_{pre}):
exact:

$$p_{\text{new}}(n_1, n_2) = p_{\text{pre}}(n_1, n_2) \times \frac{p_{\text{post}}[N]}{p_{\text{pre}}[N]}$$

New N (coarsely added probability):

Why L/R and not product-convolution?

Product-convolution uses marginals

$m_1(n_1) \cdot m_2(N-n_1)$ (a flat prior that ignores which reaction changed N).

Stochastic lifting $\mathcal{L}$: $N \rightarrow N+1$

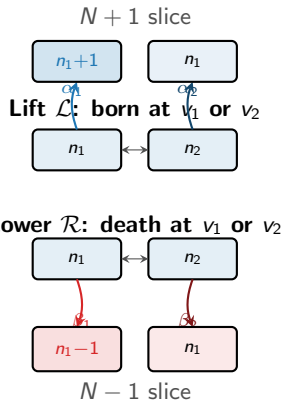
When a molecule is born, it goes to v_1 or v_2 with probabilities determined by the actual propensities:

$$\alpha_1(n_1) = \frac{b_1(n_1)}{b_1(n_1) + b_2(n_2)}, \quad b_i(n) = \alpha_{\text{rxn}}(n) + d \mu_i^{\text{ext}}$$

$$\mathcal{L}[p(\cdot | N)](n_1) = \alpha_1(n_1-1) p(n_1-1 | N) + \alpha_2(n_1) p(n_1 | N)$$

Stochastic lowering $\mathcal{R}$: $N \rightarrow N-1$

Conditioned on a death event, molecule leaves from



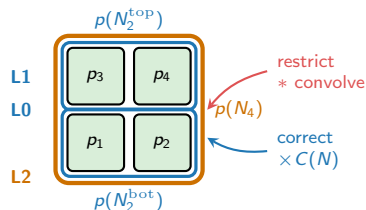
Extending to true spatial multigrid: three spatial scales

The pair V-cycle captures 2-voxel correlations.

A **full spatial multigrid** adds coarser levels:

Spatial hierarchy (active wavefront)

L0	individual voxels	w states
L1	adjacent pairs	$p(N_2)$, 1-D
L2	2×2 quads	$p(N_4)$, 1-D



V-cycle (time split:

$$\alpha_{\text{pre}} + (1-2\alpha) + \alpha_{\text{post}} = \Delta t):$$

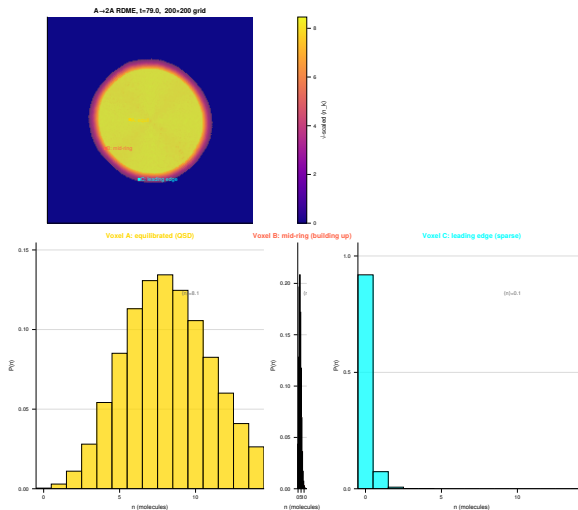
- 1 Pre-smooth L0: per-voxel expv for $\alpha\Delta t$
- 2 Restrict L0 $\rightarrow$ L1: $p_{12}(N_2) = p_1 * p_2$ (convolution)
- 3 Restrict L1 $\rightarrow$ L2: $p_{1234}(N_4) = p_{12} * p_{34}$
- 4 **Coarse solve L2**: 1-D CME for $(1-2\alpha)\Delta t$
- 5 Correct L2 $\rightarrow$ L1:

Multiplicative correction at each level:

$$p_k^{\text{new}}(n) \propto p_k(n) \times \sum_N \frac{p^{\text{new}}[N]}{p^{\text{old}}[N]} p_{\text{other}}(N-n)$$

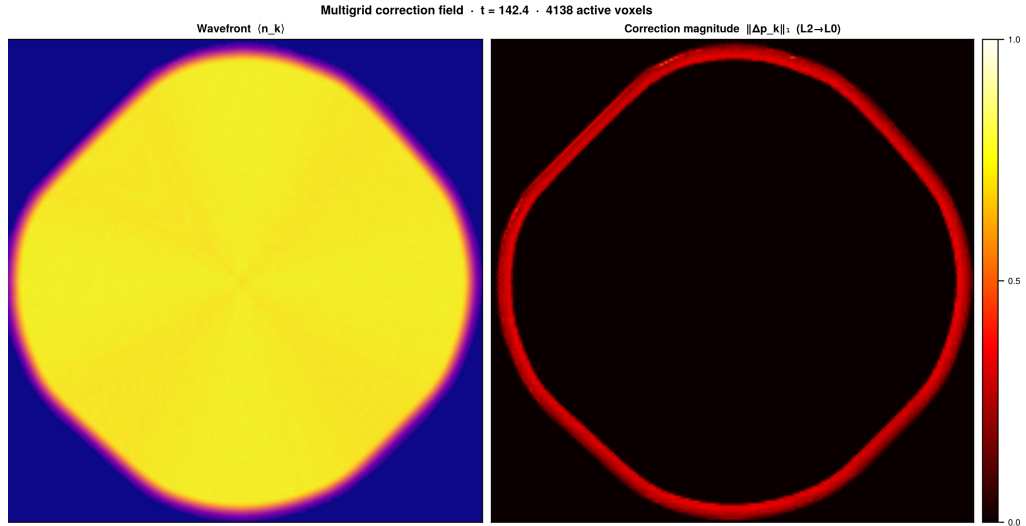
Exact for existing N ; identity fallback for new N .

Result: multigrid RDME wavefront on 200×200 grid



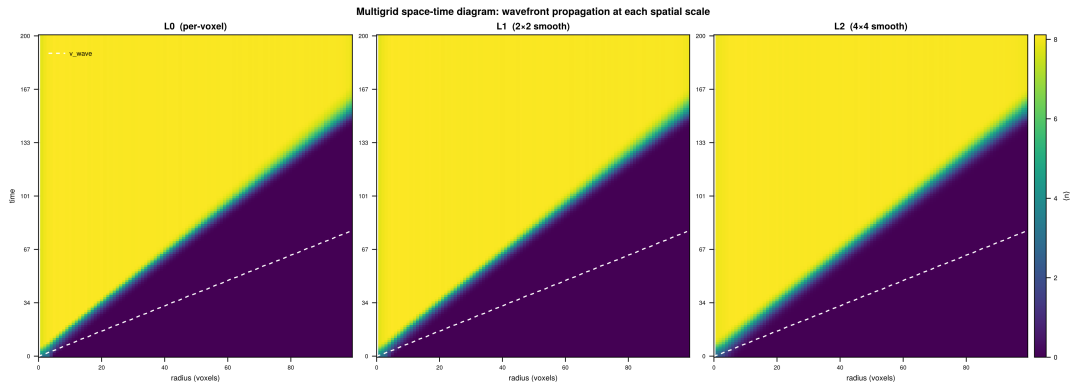
Branching-death RDME ($k_b=0.5$, $k_d=0.1$, $D=1.0$), 200×200 , three-level multigrid (L0+L1+L2). **Top**: mean-field heatmap $\langle n_k \rangle$ at $t=79.2$. **Bottom**: per-voxel distributions at three positions: interior (broad Poisson), mid-ring (sharp,

The correction field: where does L2 add information?



At peak wavefront ($t=142$. 4138 active voxels. 200×200 grid).

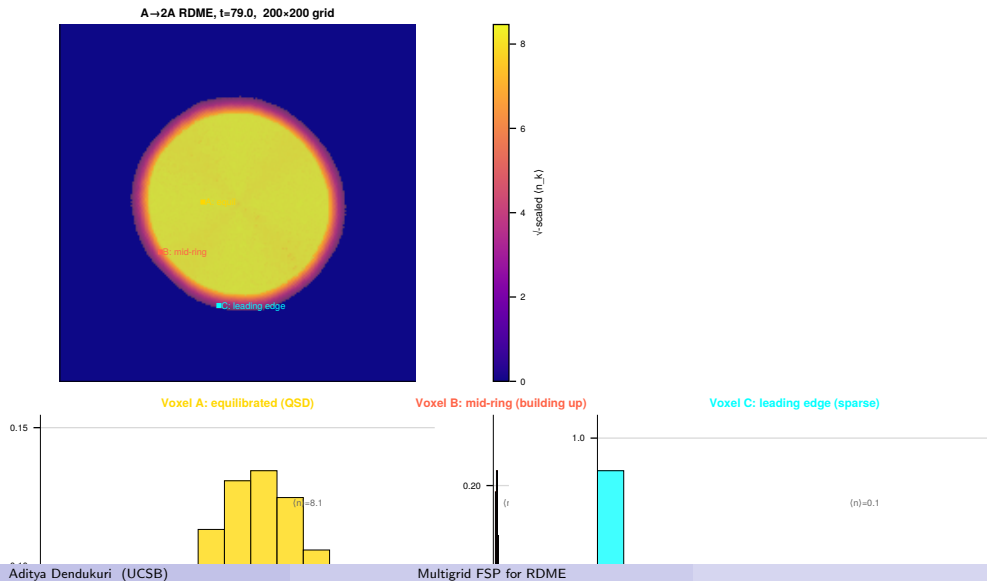
Space-time diagram: wavefront at each spatial scale



Each panel: heatmap of radial mean $\langle n \rangle$ vs. radius (x) and time (y). White dashed: theoretical wave speed $v \approx 1.26$ vox/s. **L0** (per-voxel): sharp front, thin band. **L1** (2×2 smooth): slightly broader — pairs see a wider front. **L2** (4×4 smooth): broadest — quads capture collective dynamics ahead of the fine-level front.

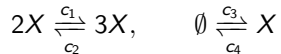
This non-redundancy across levels is why the multigrid hierarchy is effective: each level carries distinct spatial-scale information about the wave propagation.

Result: $A \rightarrow 2A$ wavefront on a 200×200 RDME



Bistable Schlögl RDME: two stable basins per voxel

Schlögl model:

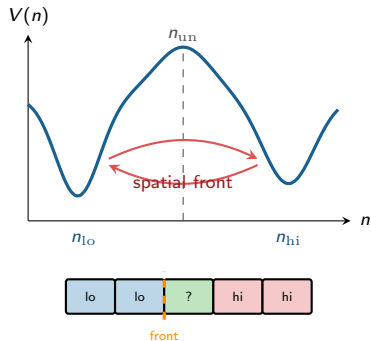


Two stable fixed points per voxel: $n_{lo} \approx 38$, $n_{hi} \approx 164$, unstable point $n_{un} \approx 63$.

Why mean-field still works

At the front, equil voxels are sharp (delta-like at n_{lo} or n_{hi}), so μ_ℓ fully describes them — mean-field is **exact** for equil-to-active coupling.

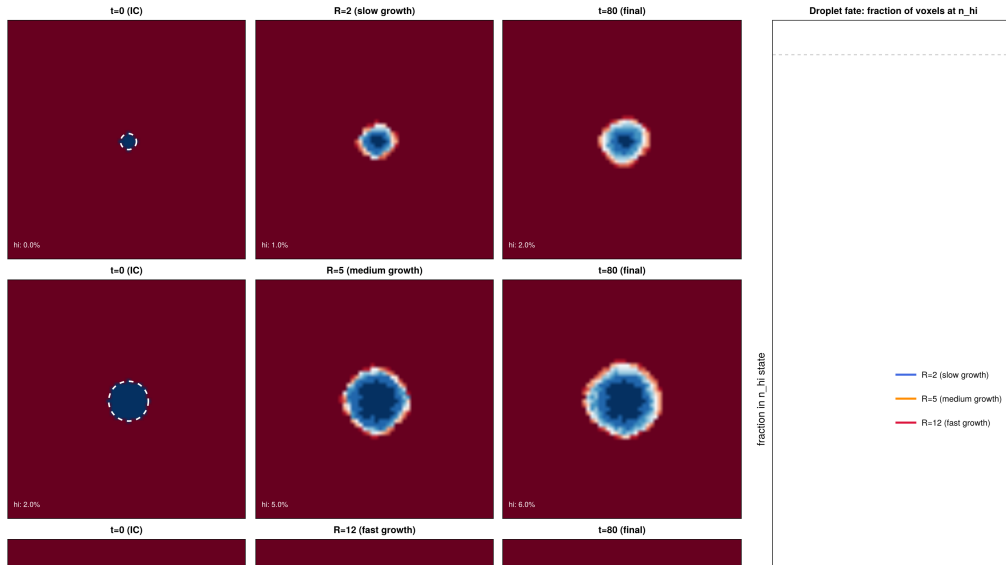
Active-active coupling handled by the ADI pair V-cycle exactly.



$P(hi)$ colormap

Color each voxel by $P(n > n_{un})$: 0 (equil-lo) $\rightarrow$

Result: Schlögl critical droplet on a 60×60 grid



Summary

1. Per-voxel CME with mean-field coupling.

The full RDME is never assembled. Diffusion is approximated by a Poisson in-flux at the neighbor mean — exact for sharp (equil) voxels. Each voxel evolves independently via `expv` on a compact tridiagonal generator.

2. ADI pair V-cycle: exact 2-voxel RDME at the front.

Adjacent active voxel pairs solved exactly (no mean-field between them). Row/column alternation (ADI) gives isotropic 2-D propagation. Pre/post-smooth equilibrate the intra-pair split; a 1-D coarse CME captures the total-count dynamics with external boundary flux.

3. Multi-level stochastic prolongation.

Three-level hierarchy on the split axis. Existing N : exact multiplicative correction. New N : physics-derived $\mathcal{L}/\mathcal{R}$ operators propagate the conditional fiber. Strictly better than product-convolution for bistable systems.